Name_____ Date_____ Period_____

**LESSON 7-4 · HOMEWORK PACKET · SHEET 1 OF 2**

# Comparing Numbers on a Number Line

Maryland Standard 6.NS.C.7a · Learning target – I can compare two rational numbers using their positions on a number line.

**REMEMBER**

- Numbers get **greater to the right** and **less to the left**.
- Any positive number is greater than any negative number.
- $-2 > -8$ because $-2$ is farther right. **<** means is less than; **>** means is greater than.

**WORKED EXAMPLE**

Compare $-3.5$ and $-1.5$. Use < or >.

**Step 1** $-1.5$ is to the right of $-3.5$ on the number line.

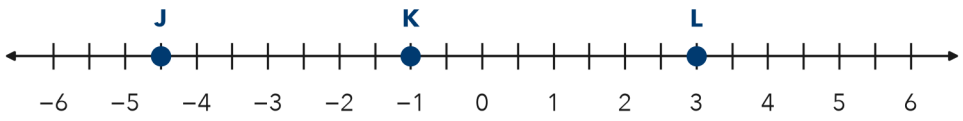
**Step 2** Right means greater.

**Answer** **$-1.5 > -3.5$**

## PRACTICE

*show your work · label units*

**1.** Use the number line. Write < or > in each blank.



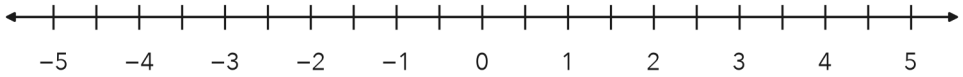

**a.** J ___ K **b.** L ___ K **c.** J ___ L

**2.** Write < or > in each blank.

**a.** $-8$ ___ $-3$ **b.** 0 ___ $-6$ **c.** $-2.5$ ___ $-2.05$

**d.** 4 ___ $-10$ **e.** $-1/2$ ___ $-3/4$ **f.** $-7$ ___ $-7.1$

**3.** Plot $-3.5$ and $-0.5$ on the number line. Then write a true comparison using <. _____




**4.** **Golf.** In golf, the lower score wins. Tanya scored $-3$. Malik scored $-5$.

**a.** Compare the scores using < or >. _____

**b.** Who won? _____

**5.** **Cold Days.** Monday's low was $-7$ degrees. Tuesday's low was $-2$ degrees. Which day was colder? Write a comparison using < or >.

**Answer**_____

**6.** **Reasoning. Is $-6 < |-6|$ true? Explain how you know.**
**Sentence Frame: "I know this because..."**

_____

_____

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** Ask someone at home: which is colder, $-4$ degrees or $-9$ degrees? Why?

Name_____ Date_____ Period_____

**LESSON 7-4 · HOMEWORK PACKET · SHEET 2 OF 2**

# Ordering Integers and Rational Numbers

Maryland Standard 6.NS.C.7b · Learning target – I can order integers and rational numbers from least to greatest.

**REMEMBER**

- Place the numbers on a number line. **Least** is farthest left; **greatest** is farthest right.
- Change fractions to decimals to compare: $-3/4 = -0.75$.
- For negative numbers, the one farther from 0 is **less**.

**WORKED EXAMPLE**

Order $-1.2$, $0.5$, $-2$, and $-1/2$ from least to greatest.

**Step 1** Write $-1/2$ as $-0.5$.

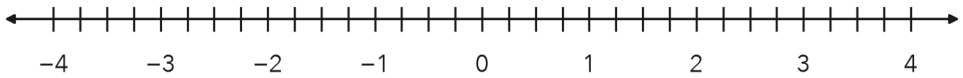
**Step 2** Read the number line from left to right.

**Answer** **$-2, -1.2, -1/2, 0.5$**

## PRACTICE

*show your work · label units*

**1.** Plot each number. Then write them in order from least to greatest: 1.5, −2.5, −3/4, 2 1/4, −3.25




**Order**_____

**2.** Order each set from least to greatest.

**a.** −6, 4, −9, 0 _____

**b.** −0.3, −3, −0.03 _____

**3.** Circle the greatest number in each set.

**a.** −5, −2, −8

**b.** −1.4, −1.04, −1.44

**c.** −2/3, −1/2, −3/4

**d.** 0, −0.1, −10

**4.** **Ocean Animals.** A shark swims at −45.5 m, a turtle at −12 m, a dolphin at −30.25 m, and a jellyfish at −8.75 m. List the animals from deepest to closest to the surface.

**Answer**_____

**5.** **Find the Error.** Leo ordered −1/2, −0.4, and −0.6 from least to greatest as −0.4, −1/2, −0.6. Explain his mistake and write the correct order.

_____

_____

## TEST PRACTICE

*choose the best answer*

**6.** Which list is ordered from least to greatest?

**A** −5, −8, 0, 3 **B** −8, −5, 0, 3 **C** 0, −5, −8, 3 **D** 3, 0, −5, −8

**7.** Which number is greatest?

**A** −2.7 **B** −2.07 **C** −2.77 **D** −2.2

**CHECK YOURSELF** ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

**AT HOME** List the low temperatures for three cold days with someone at home. Order them from coldest to warmest.